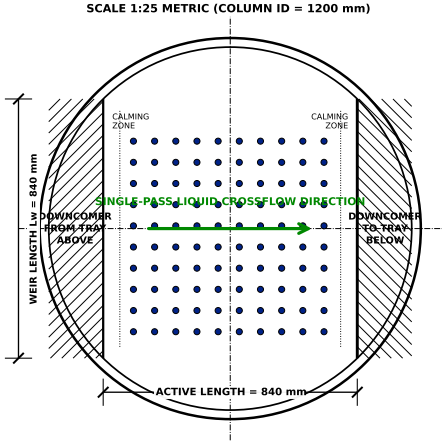
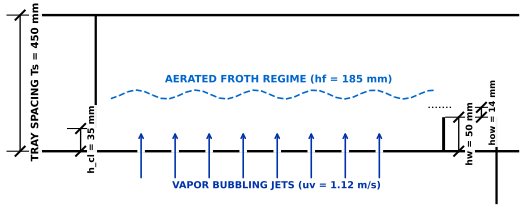
VIEW A: SINGLE-PASS SIEVE TRAY PLAN VIEW

SCALE 1:25 METRIC (COLUMN ID = 1200 mm)

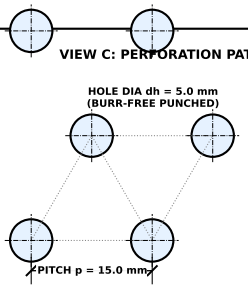


VIEW B: SECTION B-B — TRAY FROTH HYDRAULICS

DETAIL OF CROSSFLOW FROTH REGIME & DOWNCOMER BACKUP



VIEW C: PERFORATION PATTERN & EQUILATERAL TRIANGULAR PITCH



PERFORATION SPECIFICATIONS:
- Hole Diameter (dh): 5.0 mm (3/16") punched 304L SS sheet
- Pitch (p): 15.0 mm Equilateral Triangular (30 deg orientation)
- Hole-to-Active Area Ratio (Ah/Aa): 10.1 % [Ah = 0.089 m2]
- Total Perforation Holes: 4,530 Holes per tray floor
- Sheet Thickness: 2.5 mm (12 B.W.G.)
- Fabrication: Precision CNC gang-punched, smooth deburred top

TABLE 1: SIEVE TRAY GEOMETRY & PHYSICAL DIMENSIONS

GEOMETRIC PARAMETER	SYMBOL	VALUE	UNIT
Column Internal Diameter	Dc	1200	mm
Total Tower Cross-Section Area	At	1.131	m2
Downcomer Area (Inlet / Outlet)	Ad	0.124 (11.0 %)	m2
Net Free Area (At - Ad)	An	1.007	m2
Active Bubbling Area	Aa	0.883 (78.1 %)	m2
Total Perforation Hole Area	Ah	0.089 (10.1 % Aa)	m2
Weir Length (Segmental Chord)	Lw	840	mm
Weir Height	hw	50.0	mm
Downcomer Clearance Height	hcl	35.0	mm
Tray Spacing	Ts	450	mm

TABLE 2: HYDRAULIC VALIDATION & PRESSURE DROP BREAKDOWN

HYDRAULIC METRIC	CALCULATED	ALLOWABLE	STATUS / CHECK
Operating Vapor Velocity (uv)	1.12 m/s	1.64 m/s (uf)	PASS (68.3 % Flooding)
Dry Tray Pressure Drop (hd)	28.4 mm liq	< 40.0 mm	PASS (Orifice drop)
Liquid Height Over Weir (how)	14.2 mm liq	> 10.0 mm	PASS (Francis Formula)
Residual Surface Tension Drop (hr)	9.8 mm liq	5 - 15 mm	PASS (Aeration drop)
Total Tray Pressure Drop (ht)	62.4 mm (540 Pa)	< 75 mm (650 Pa)	PASS (Optimal range)
Downcomer Head Loss (hdc)	8.6 mm liq	< 25.0 mm	PASS (Sub-weir flow)
Clear Downcomer Backup (hdc_tot)	165 mm liq	< 225 mm (50 % Ts)	PASS (Zero choke flood)
Fractional Entrainment (psi)	0.021 kg/kg	< 0.050 kg/kg	PASS (Low entrainment)

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
SIEVE TRAY HYDRAULIC DESIGN & INTERNAL ASSEMBLY	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	AS NOTED
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/ED/DWG-04	01 / 09 / 2026

ETHANOL DISTILLATION (C-201) | 700 kmol/h FEED (7 mol% EtOH) | ID 1.20 m, R=3.0 | ASME SEC VIII / TEMA